Algebra Monsoon 2026 Lecture 7

Categories and functors

https://simonwillerton.staff.shef.ac.uk/TheCatsters/

https://math.ucr.edu/home/baez/act_course/index.html

A quiver Q is a quadruple

$$Q = (E_0, E_1, s_0, s_1)$$

consisting of a set E_0 (whose elements are called the vertices of Q) a set E_1 (whose elements are called the arrows of Q) and a mapping

$$(s_0, s_1) : E_1 \longrightarrow E_0 \times E_0$$

taking each arrow $f \in E_1$ to its source $s_0(f)$ and its target $s_1(f)$.

For $f \in E_1$, we write $f : A \longrightarrow B$ to say that $s_0(f) = A$ and $s_1(f) = B$.

We denote by

$$\mathrm{Hom}_{\mathcal{Q}}(A, B) = (s_0, s_1)^{*}(A, B)$$

the set of such arrows.

- A pair of arrows (f, g) is said to be composable if

$$s_1(f) = s_0(g).$$

Examples $\emptyset$, $\cdot$, $\circlearrowright$,

etc.

For any relation $R \subset E \times E$, take $Q_0 = E$, $Q_1 = R$, s_0 = first proj., s_1 = second proj.

- The set of composable pairs of arrows is $E_1 \underset{(s_1, s_0)}{\times} E_1$ (denoted $E_1 \underset{E_0}{\times} E_1$)

$= \{(f, g) : s_1(f) = s_0(g)\}$.

Def A morphism of quivers from $Q = (E_0, E_1, s_0, s_1)$ to $Q' = (E_0', E_1', s_0', s_1')$ is a pair of mappings $f_0 : E_0 \to E_0'$, $f_1 : E_1 \to E_1'$ such that the square

$$\begin{array}{ccc} E_1 & \xrightarrow{f_1} & E_1' \\ {\scriptstyle (s_0, s_1)}\downarrow & & \downarrow{\scriptstyle (s_0', s_1')} \\ E_0 \times E_0 & \xrightarrow{f_0 \times f_0} & E_0' \times E_0' \end{array}$$

commutes.

Ol [illegible] $\subset E \times E_1$ then $(f(a), f(e)) \in E_1' \underset{E_0'}{\times} E$

- It follows that if $(a,b) \in E_1 \underset{E_0}{\times} E_1$, then $(f(a), f(b)) \in E_1' \underset{E_0'}{\times} E_1'$

A category $\mathcal{C}$ is a quiver

$$(E_0, E_1, s_0, s_1)$$

together with mappings

$$Id : E_0 \longrightarrow E_1$$

all $(f,g) \in E_1 \times E_1$ such that $s_1(f) = s_0(g)$

$$\circ : E_1 \underset{(s_1, s_0)}{\times} E_1 \longrightarrow E_1$$

$$(f,g) \longmapsto g \circ f$$

such that

1) $s_0(Id_A) = s_1(Id_A) = A$

for every $A \in E_0$.

$$s_0(g \circ f) = s_0(f)$$

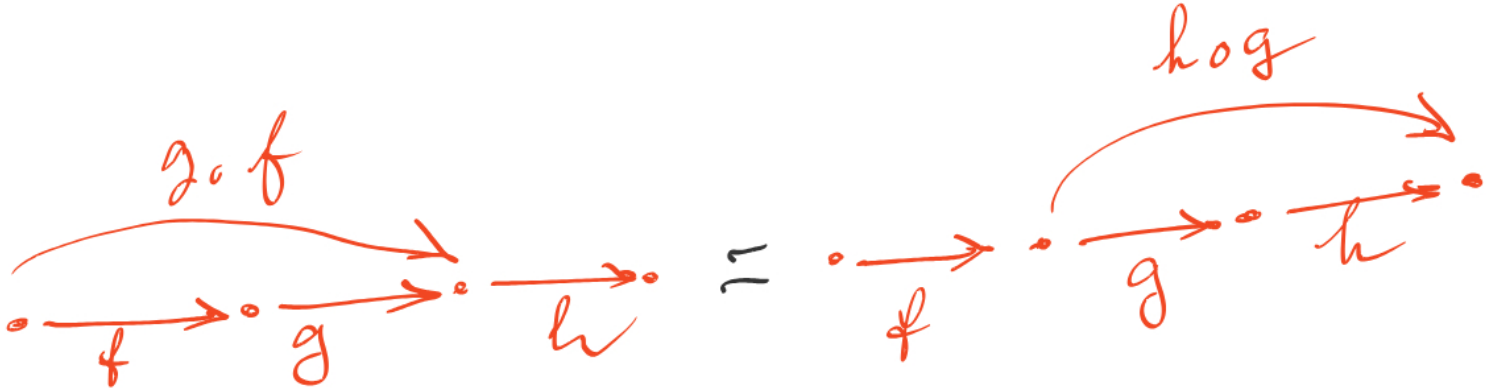
$$s_1(g \circ f) = s_1(g)$$

for every pair $(f,g) \in E_1 \underset{(s_1, s_0)}{\times} E_1$.

1) $Id_B \circ f = f$ and $f \circ Id_A = f$

for every $f \in Hom(A, B)$.

2) $h \circ (g \circ f) = (h \circ g) \circ f$ for any triple of morphisms $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$ -



In a category, vertices are called objects and arrows are called morphisms.

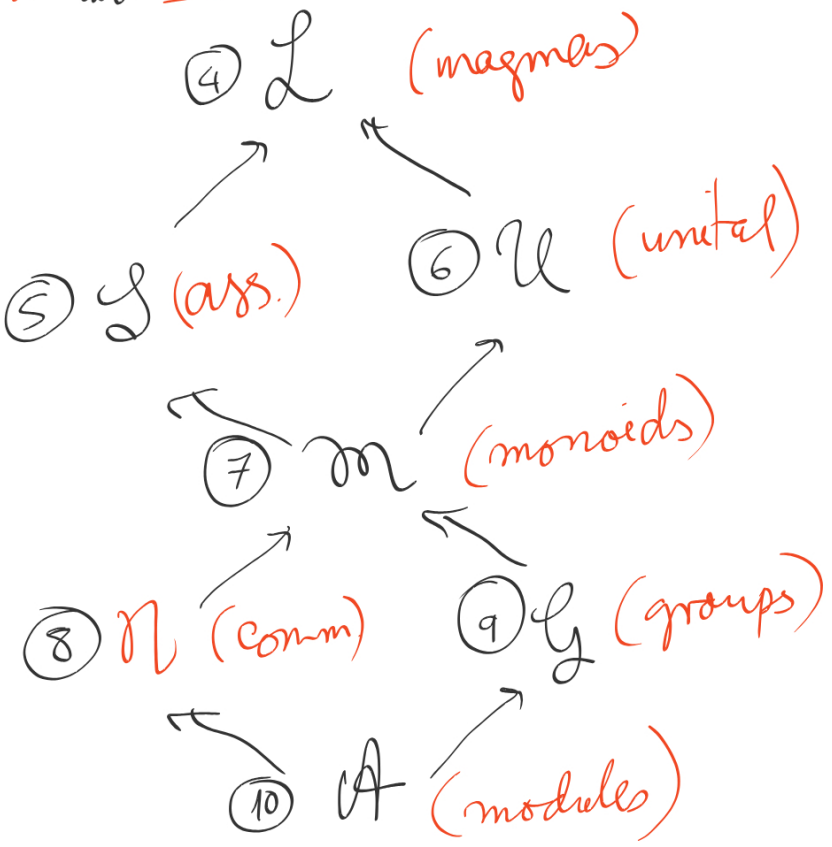
Examples

(3) Quivers and their morphisms

(0) $\emptyset$

(1) $\mathcal{E}$: Sets and mappings

(2) Sets and correspondences



③ $\mathcal{L}$: magmas and morphisms

⑯ $\mathcal{A}$: modules and morphisms

⑪ • Various kind of pairs, triples, we have considered earlier.

⑫ • Let $(S, \leq)$ be an ~~o~~ordered set

objects = elements of S

For $a, b \in S$, there is a unique morphism if $a \leq b$, otherwise $\mathrm{Hom}(a, b) = \emptyset$.

⑬ • For any monoid M, we have a category $\mathcal{BM}$ with just one object and with M as the set of morphisms.

Let $\mathcal{C}$ be a category

Def We say that $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ is an isomorphism if there exists a $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$ such that

$$g \circ f = Id_A \quad \text{and} \quad f \circ g = Id_B.$$

- $f = Id_A$ is an isomorphism: take $g = Id_A$.

- If g exists, it is unique. We write $g = f^{-1}$.

Indeed, if g_1, g_2 are two such morphisms (so that

$$g_1 \circ f = Id_A \quad , \quad g_2 \circ f = Id_A$$
$$f \circ g_1 = Id_B \quad , \quad f \circ g_2 = Id_B \Big)$$

then

$$\begin{aligned} g_2 &= Id_A \circ g_2 \\ &= (g_1 \circ f) \circ g_2 \\ &= g_1 \circ (f \circ g_2) \\ &= g_1 \circ Id_B \\ &= g_1 \end{aligned}$$

□

- If f is an isomorphism, then so is f^{-1} and $(f^{-1})^{-1} = f$.

- If $f \in \mathrm{Hom}_{\mathcal{C}}(A,B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B,C)$ are isomorphisms, then so is
$$g \circ f \in \mathrm{Hom}_{\mathcal{C}}(A,C)$$
and $$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Indeed
$$\begin{aligned}(g \circ f) \circ (f^{-1} \circ g^{-1}) &= g \circ (f \circ f^{-1}) \circ g^{-1} \\ &= g \circ Id_A \circ g^{-1} \\ &= g \circ g^{-1} \\ &= Id_B\end{aligned}$$

and similarly
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = Id_A.$$

Therefore $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

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$\mathrm{Hom}(A,A)$ is a monoid

Therefore $(g \circ f)$ — 0 0

- For every object A, $\mathrm{End}_{\mathcal{C}}(A) = \mathrm{Hom}_{\mathcal{C}}(A, A)$ is a monoid and $\mathrm{Aut}_{\mathcal{C}}(A) = \mathrm{End}_{\mathcal{C}}(A)^{\times}$ is a group.

Elements of $\mathrm{End}_{\mathcal{C}}(A)$ are called $\mathcal{C}$-endomorphisms of A

$\mathrm{Aut}_{\mathcal{C}}(A)$ —— $\mathcal{C}$-automorphisms ——.

Question: What are the isomorphisms in the category of sets and correspondences?

Answer: Precisely the (graphs of) bijections.

Def A morphism f is said to be a monomorphism if it is left cancellable ($fg_1 = fg_2 \Longrightarrow g_1 = g_2$

Equivalently, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ is a monomorphism if for every object C of $\mathcal{C}$, the mapping

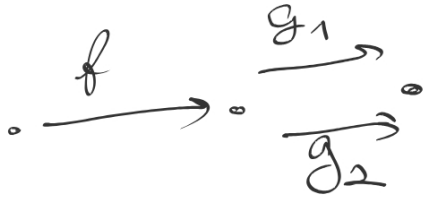
$$\mathrm{Hom}_{\mathcal{C}}(C, A) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(C, B)$$
$$g \longmapsto f \circ g$$

is injective.

(In the category $\mathcal{C}$ of divisible modules, the morphism $\mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z}$ is a monomorphism (exercise) but not injective.)

- In the category of sets, a morphism is a monomorphism if and only if it is injective.

Def A morphism f is said to be an epimorphism if it is right cancellable ($g_1 f = g_2 f \Rightarrow g_1 = g_2$)



Equivalently, a morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ is an epimorphism if for every object C of $\mathcal{C}$, the mapping

$$\mathrm{Hom}_{\mathcal{C}}(B, C) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$$
$$g \longmapsto g \circ f$$

is injective

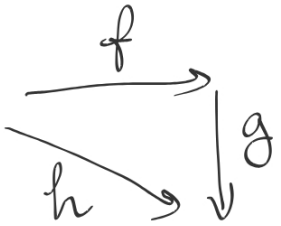
Exercise Show that the inclusion $i : \mathbb{Z} \to \mathbb{Q}$ in the category of commutative monoids is an epimorphism.

- In the category of sets, a morphism is an epimorphism if and only if it is a surjective mapping.

Exercise An isomorphism is both a monomorphism and an epimorphism.

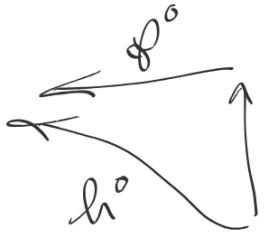
Def Let $\mathcal{C}$ be a category. The opposite category $\mathcal{C}^\circ$ has the same set of objects but $\mathrm{Hom}_{\mathcal{C}^\circ}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A)$.

(if



commutes in $\mathcal{C}$

then



commutes in $\mathcal{C}^\circ$.

Def Let $\mathcal{C}$ and $\mathcal{D}$ be categories.
A functor $F : \mathcal{C} \to \mathcal{D}$

this says that F is a morphism of quivers

assigns to every object A of $\mathcal{C}$ an object $F(A)$ of $\mathcal{D}$ and

to every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ a morphism $F(f) \in \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$

in such a way that

- $F(\mathrm{Id}_A) = \mathrm{Id}_{F(A)}$ for all $A \in \mathcal{C}_0$
- $F(g \circ f) = F(g) \circ F(f)$ for all $(f, g) \in \mathcal{C}_1 \times_{\mathcal{C}_0} \mathcal{C}_1$

It follows that F induces a morphism of monoids $\mathrm{End}_{\mathcal{C}}(A) \to \mathrm{End}_{\mathcal{D}}(F(A))$ for every $A \in \mathcal{C}_0$.

$\mathrm{End}_{\mathcal{C}}(A)$ — endomorphisms of an object X of a category $\mathcal{C}$,

It follows that F induces a [illegible] for every $A \in \mathcal{C}_0$:
$\mathrm{End}_{\mathcal{C}}(A) \longrightarrow \mathrm{End}_{\mathcal{D}}(F(A))$

Example For every object X of a category $\mathcal{C}$, we have the functor

$$h^X : \mathcal{C} \longrightarrow \mathcal{E}$$

defined by $h^X(Y) = \mathrm{Hom}_{\mathcal{C}}(X, Y)$.

for every object Y of $\mathcal{C}$. and

$$h^X(f) = f \circ g \quad (g \in \mathrm{Hom}_{\mathcal{C}}(X, Y))$$

for every morphism $f : Y \to Z$ in $\mathcal{C}$.

- Forgetful functors $\mathcal{L} \to \mathcal{E}$, $\mathcal{G} \to \mathcal{E}$ etc.

- Free object functors $F_{\mathcal{L}}$:

Let M be a monoid. View M as a category $B(M)$ such that $E_0 = \{M\}$, $E_1 = M$

$\mathcal{B}(M)$

Exercise Let M_1, M_2 be monoids. Show that for every functor $F : \mathcal{B}(M_1) \to \mathcal{B}(M_2)$, the resulting mapping $\varphi(F) : M_1 \to M_2$ is a morphism of monoids and that φ is a bijection

$$[\mathcal{B}(M_1), \mathcal{B}(M_2)] \longrightarrow \mathrm{Hom}_m(M_1, M_2)$$

- Let M be a monoid. A functor $\mathcal{B}(M) \to \mathcal{E}$ is nothing but an M-set (as we shall see)

A functor $\mathcal{B}(G) \to \mathcal{A}$ is nothing but a G-module

- $\mathcal{B}$ is a functor from the category of monoids to the category of categories.

Example For every object X of a category $\mathcal{C}$, we have a functor

$$h_X : \mathcal{C} \longrightarrow \mathcal{E}^{o} \quad (\mathcal{E} = \text{sets and mappings})$$

defined by

$$h_X(Y) = \mathrm{Hom}_{\mathcal{C}}(Y, X)$$

for every object Y of $\mathcal{C}$ and

for every morphism $\phi : Y \longrightarrow Z$ in $\mathcal{C}$,

$$h_X(\phi) = (g \mapsto g \circ \phi) \qquad \left(g \in \mathrm{Hom}_{\mathcal{C}}(Z, X)\right)$$

Exercise Functors take isomorphisms to isomorphisms.

(Meaning: If $F: \mathcal{C} \to \mathcal{D}$ is a functor and $\alpha \in \mathrm{Hom}_{\mathcal{C}}(A,B)$ is an isomorphism, then so is $F(\alpha) \in \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$

- For every category $\mathcal{C}$, we have the functor

$$\mathrm{Id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}.$$

(defined by $\mathrm{Id}_{\mathcal{C}}(X) = X$ for every object X of $\mathcal{C}$ and $\mathrm{Id}_{\mathcal{C}}(f) = f$ for every morphism f in $\mathcal{C}$).

- If $F: \mathcal{B} \to \mathcal{C}$ and $G: \mathcal{C} \to \mathcal{D}$ are functors, then so is $G \circ F$.

So one can consider the category whose

- objects are categories, and
- morphisms are functors.

This gives rise to the notion of isomorphism of categories:

a functor $F : \mathcal{C} \to \mathcal{D}$ for which there exists a functor $G : \mathcal{D} \to \mathcal{C}$ such that $G \circ F = Id_{\mathcal{C}}$, $F \circ G = Id_{\mathcal{D}}$

A better notion is that of equivalence of categories (just as isomorphism of groups is a better notion than equality of groups.)

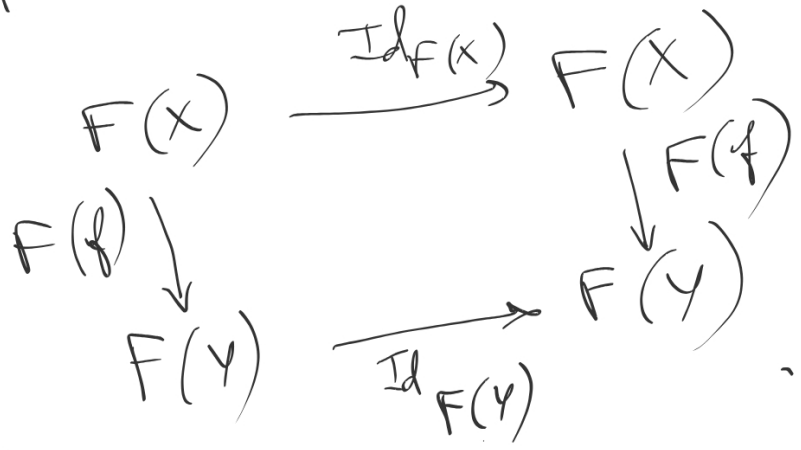
Def Let $\mathcal{C}, \mathcal{D}$ be categories and $F, G : \mathcal{C} \to \mathcal{D}$ functors.

A morphism of functors (or a functorial morphism) $\eta : F \to G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ in $\mathcal{D}$ indexed by objects X of $\mathcal{C}$ such that for every morphism $f : X \to Y$ in $\mathcal{C}$, the square

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array}$$

Commutes ($\eta_Y \circ F(f) = G(f) \circ \eta_X$)

- For every functor $F: \mathcal{C} \to \mathcal{D}$, we have the morphism of functors Id_F defined by $(Id_F)_X = Id_{F(X)}$ for every object X of $\mathcal{C}$. Note that for every morphism $f: X \to Y$ in $\mathcal{C}$, the square



commutes

- If $F, G, H : \mathcal{C} \to \mathcal{D}$ are functors and $\xi : F \to G$, $\eta : G \to H$ are morphisms of functors, then so is $\eta \circ \xi$, defined by

$$(\eta \circ \xi)_X = \eta_X \circ \xi_X$$

for every object X of $\mathcal{C}$ because for every morphism $f : X \to Y$ in $\mathcal{C}$, the rectangle

$$\begin{array}{ccccc} F(X) & \xrightarrow{\xi_X} & G(X) & \xrightarrow{\eta_X} & H(X) \\ {\scriptstyle F(f)}\downarrow & & {\scriptstyle G(f)}\downarrow & & {\scriptstyle H(f)}\downarrow \\ F(Y) & \xrightarrow[\xi_Y]{} & G(Y) & \xrightarrow[\eta_Y]{} & H(Y) \end{array}$$

commutes.

Proposition Let $\mathcal{C}$ and $\mathcal{D}$ be categories. There is a category $[\mathcal{C}, \mathcal{D}]$ whose objects are the functors $\mathcal{C} \to \mathcal{D}$ and whose morphisms are morphisms of functors defined above. A morphism $\eta : F \to G$ in $[\mathcal{C}, \mathcal{D}]$ is an isomorphism if and only if $\eta_X : F(X) \to G(X)$ is an isomorphism for every object X of $\mathcal{C}$.

Def A functor $F : \mathcal{C} \to \mathcal{D}$ is an *equivalence of categories* if there is a functor $G : \mathcal{D} \to \mathcal{C}$ such that $G \circ F$ is isomorphic to $Id_{\mathcal{C}}$ and $F \circ G$ is isomorphic to $Id_{\mathcal{D}}$. (in the categories $[\mathcal{C}, \mathcal{C}]$ and $[\mathcal{D}, \mathcal{D}]$ resp.)

(Example The category V of finite-dimensional vector spaces over a commutative field K is equivalent to the category $\mathcal{W}$ whose set of objects is $\mathbb{N}$ and $\mathrm{Hom}_{\mathcal{W}}(m, n) =$ matrices with m columns and n rows over K.)

Theorem (Yoneda) Let $\mathcal{C}$ be a category, X an object of $\mathcal{C}$ and $T : \mathcal{C} \to \mathcal{E}$ a functor.

The mapping

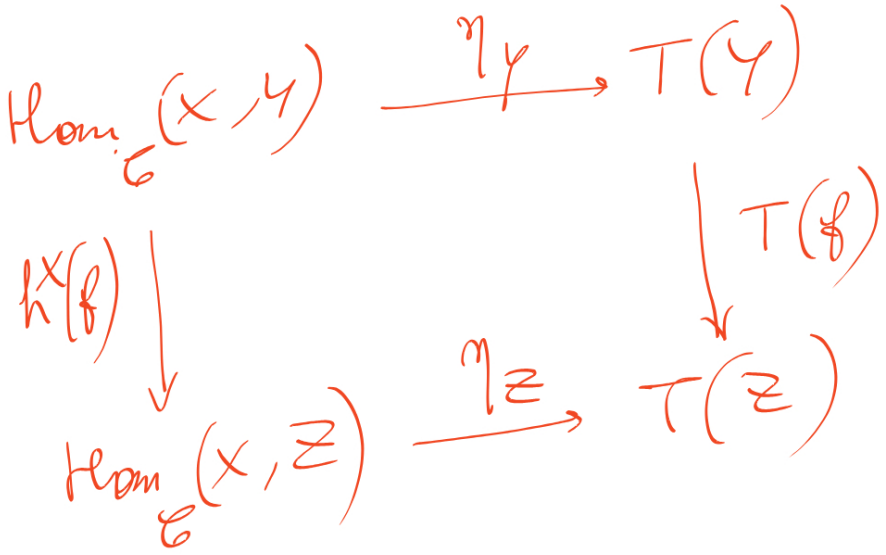
$$\begin{aligned} \mathrm{Hom}_{[\mathcal{C},\mathcal{E}]}(h^X, T) &\xrightarrow{\sim} T(X) \\ \varphi : \eta &\longmapsto \eta_X(\mathrm{Id}_X) \end{aligned}$$

is a bijection.

In particular, for $T = h^Y$, we have the bijection $\mathrm{Hom}_{[\mathcal{C},\mathcal{E}]}(h^X, h^Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(Y, X)$

and, for every morphism in $\mathcal{C}$ (resp. $[\mathcal{C}, \mathcal{E}]$) the corresponding squares commute.

Proof For every $f: Y \to Z$ in $\mathcal{C}$, the square



commutes (by definition of a morphism of functors). If we take $Y = X$ and consider $\mathrm{Id}_X \in \mathrm{Hom}_{\mathcal{C}}(X,X)$, then

$$\eta_Z\left(h^X(Id_X)\right) = T(f)\left(\eta_X(Id_X)\right)$$

$$\| $$

$$\eta_Z(f \circ Id_X)$$

$$\| $$

$$\eta_Z(f)$$

So, for every object Z of $\mathcal{C}$, η_Z is uniquely uniquely determined by $\eta_X(Id_X)$.
Therefore the mapping φ is injective.

On the other hand, for $x \in T(X)$, putting $\eta_Z(f) = T(f)(x)$ defines a morphism of functors $h^X \to T$ such that $h^X(Id_X) = x$ because $T(Id_X) = Id_{T(X)}$. Therefore φ is surjective. □

- The bijection $\mathrm{Hom}_{[\mathcal{C}, \mathcal{E}]}(h^X, T) \xrightarrow{\sim} T(X)$ is functorial in X and in T.

Initial objects, final objects

Let $\mathcal{C}$ be a category. An object A of $\mathcal{C}$ is called an initial object if for every object X of $\mathcal{C}$, there is a unique morphism $A \to X$ in $\mathcal{C}$. An initial object in $\mathcal{C}^{op}$ is called a final object in $\mathcal{C}$.

Example. • $\emptyset$ is an initial object in $\mathcal{E}$.

• every singleton is a final object in $\mathcal{E}$.

• $\{\ast\}$ is an initial object in $\mathcal{U}$ (unital magmas)
(also a final object)

• $\mathbb{N}$ is an initial object in
pointed $\mathcal{M}$ (monoids) and
pointed $\mathcal{N}$ (commutative monoids)

• $\mathbb{Z}$ is an initial object in
pointed $\mathcal{G}$ (groups) and
pointed $\mathcal{A}$ (modules).

• We have seen various examples
of initial objects in previous lectures.

Proposition If a category has an initial object, it is unique up to a unique isomorphism.

Proposition If a category has a final object, it is unique up to a unique isomorphism.

Products and Coproducts

Let $\mathcal{C}$ be a category and $(A_\lambda)_{\lambda \in \Lambda}$ a family of objects of $\mathcal{C}$. Consider

pairs $\quad (P, (\phi_\lambda)_{\lambda \in \Lambda})$

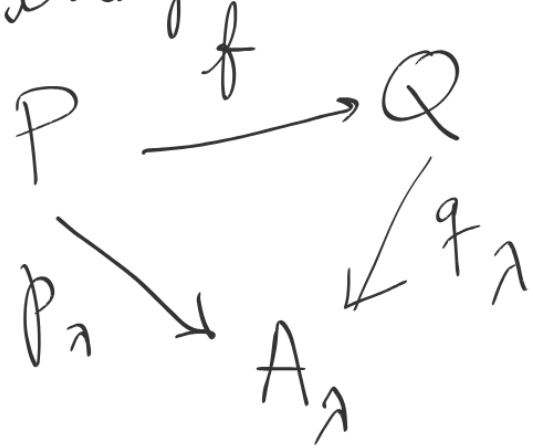
consisting of an object P of $\mathcal{C}$ and morphisms $\phi_\lambda \in \mathrm{Hom}_{\mathcal{C}}(P, A_\lambda)$.

Define a morphism of pairs

$$\left(P, (p_\lambda)_{\lambda \in \Lambda}\right) \to \left(Q, (q_\lambda)_{\lambda \in \Lambda}\right)$$

to a morphism $f \in \mathrm{Hom}_{\mathcal{C}}(P, Q)$

such that the triangle



commutes for every $\lambda \in \Lambda$.

- Id_P is an endomorphism of $\left(P, (p_\lambda)_{\lambda\in\Lambda}\right)$

- If $\left(P, (p_\lambda)_{\lambda\in\Lambda}\right) \xrightarrow{f} \left(Q, (q_\lambda)_{\lambda\in\Lambda}\right)$
$\xrightarrow{g} \left(R, (r_\lambda)_{\lambda\in\Lambda}\right)$

are morphisms of pairs, then so is

- $g\circ f : \left(P, (p_\lambda)_{\lambda\in\Lambda}\right) \to \left(R, (r_\lambda)_{\lambda\in\Lambda}\right)$

As the identity and associativity axioms hold, we are in the presence of a category. ☺

Definition A final object in the category $\mathcal{C}$ is called a product of the family $(A_\lambda)_{\lambda \in \Lambda}$ in $\mathcal{C}$.

- If the product exists, it is unique up to a unique isomorphism.

Notation $\prod_{\substack{\mathcal{C} \\ \lambda \in \Lambda}} A_\lambda$

Examples Products exist in the categories

$\mathcal{E}$ (sets and mappings)

$\mathcal{L}$ (magmas)

$\mathcal{M}$ (monoids)

$\mathcal{N}$ (commutative monoids)

$\mathcal{G}$ (groups)

$\mathcal{A}$ (modules)

etc.

Exercise Define the coproduct of a family of objects $(A_\lambda)_{\lambda \in \Lambda}$ in the category $\mathcal{C}$.

Notation $\coprod_{\lambda \in \Lambda}^{\mathcal{C}} A_\lambda$.

Example If $\mathcal{C} = \mathcal{A}$, then

$\coprod_{\lambda \in \Lambda}^{\mathcal{A}} A_\lambda$ is very different

from $\coprod_{\lambda \in \Lambda}^{\mathcal{G}} A_\lambda$ which is very different

from $\coprod_{\lambda \in \Lambda}^{\mathcal{E}} A_\lambda$.

Let $(E, \preceq)$ be an ordered set, viewed as a category. For objects $a, b \in E$

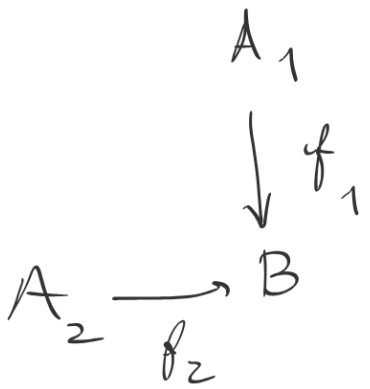
- $\mathrm{Inf}(a,b) \preceq a$ and $\mathrm{Inf}(a,b) \preceq b$
- $(c \preceq a \text{ and } c \preceq b) \Rightarrow c \preceq \mathrm{Inf}(a,b)$

Therefore $\mathrm{Inf}(a,b)$ (if it exists) is the categorical product of a and b.
(One writes $a \wedge b$ instead of $a \times b$).

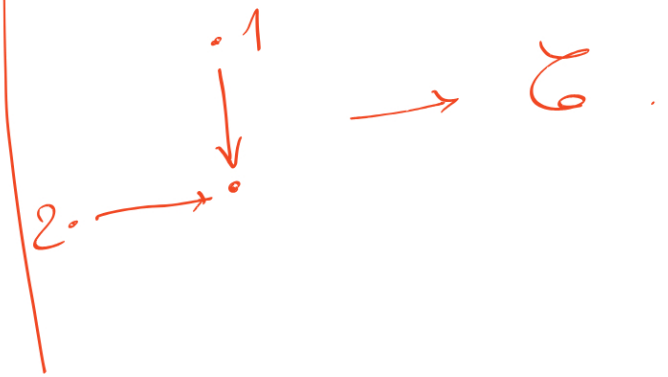
What is the coproduct $A \amalg_{\mathcal{C}} B$ in the category $\mathcal{C}$ of pointed sets.

Cartesian and cocartesian squares

Given a diagram



Or equivalently, a functor

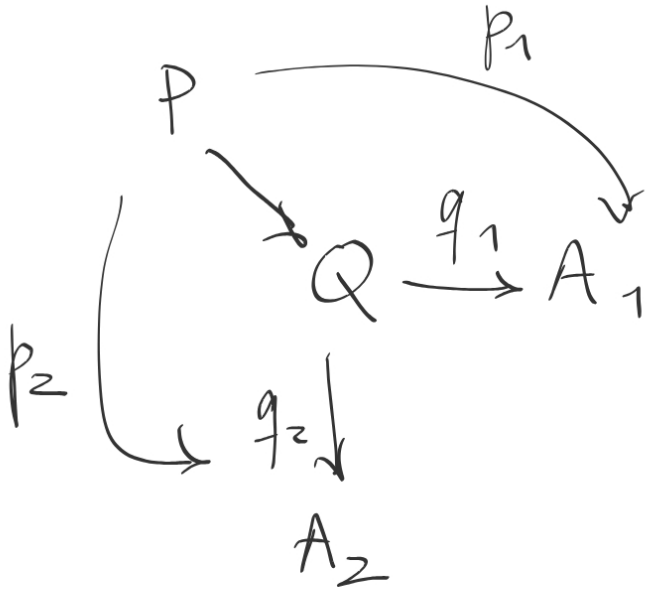


in a category $\mathcal{C}$, consider triples consisting (P, ϕ_1, ϕ_2) consisting of an object P of $\mathcal{C}$ and morphisms $\phi_1 : P \to A_1$ and $\phi_2 : A_2 \to B$ such that $f_1 \circ \phi_1 = f_2 \circ \phi_2$.

A morphism of triples

$$(P, p_1, p_2) \longrightarrow (Q, q_1, q_2)$$

is a morphism $\varphi : P \to Q$ in $\mathcal{C}$ such that the triangles



Commutes.

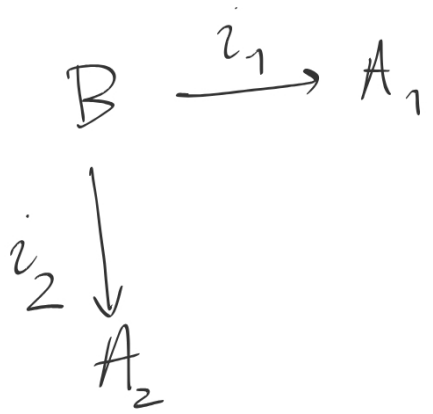
A final object in this category is called the fibre product or the limit of the diagram and denoted $A_1 \underset{(f_1, f_2)}{\times} A_2$.

Exercise

Show that in the category of sets

$$A_1 \underset{(f_1, f_2)}{\times} A_2 = \{ (a_1, a_2) \in A_1 \times A_2 \mid f_1(a_1) = f_2(a_2) \}$$

Problem By reversing arrows, define the colimit of



Exercise Show that in the category of sets, the colimit of

$$A_1 \xleftarrow{i_1} B \xrightarrow{i_2} A_2$$

is $\left(A_1 \amalg A_2\right) / \left\{ i_1(b) \sim i_2(b) \text{ for all } b \in B \right\}$.

Exercise Let (E_1, x_1), (E_2, x_2) be pointed sets. What is their coproduct in the category of pointed sets?

Exercise Let S be a set. In the category $\mathcal{P}(S)$, the square

$$\begin{array}{ccc} A \cap B & \xrightarrow{j_A} & A \\ {\scriptstyle j_B}\downarrow & & \downarrow{\scriptstyle i_A} \\ B & \xrightarrow[i_B]{} & A \cup B \end{array}$$

is both cartesian and cocartesian (ie $A \cap B$ is the limit of the two arrows i_A, i_B and $A \cup B$ is the colimit of the two arrows j_A, j_B.